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Deciphering aging-associated drivers of glioblastoma biology at the tumor core and border

Laurens Zschke^{1,2,}, Daniel Teichmann¹, Peggy Wolkenstein¹, Simone Schmid¹, Jakob Nückles¹, Christin Siewert¹, Brendan Osberg¹, Naveed Isquaaq³, Kaixin Huang³, MOSAIC Consortium⁴, Philip Dao Trong⁵, Arend Koch¹, Julia Onken¹, Matthias Endres¹, Leif Ludwig^{3,2}, David Capper¹, and Ilon Liu¹*

¹Charité - University Hospital Berlin

²Berlin Institute for Medical Systems Biology

³Berlin Institute of Health at Charité

⁴Owkin Inc.

⁵Vivantes Klinikum Neukölln

*Presenting author

Glioblastoma (GB), IDH-wildtype, is an age-related disease peaking around age 65, yet the impact of aging on its biology remains poorly understood.

Here, we present a spatially resolved multi-omics framework to characterize age-associated differences in GB. We analyzed 142 FFPE samples from 112 primary GB patients (36–86 years), including 74 patients from the MOSAIC consortium, using single-nucleus RNA sequencing, bulk DNA methylation profiling, targeted/whole-exome sequencing, and - in a subset - spatial transcriptomics. Histologically selected regions captured the full spatial heterogeneity of GB, including 30 samples from normal-appearing peritumoral brain. To establish a healthy aging reference, we profiled 16 cortical surgery samples from non-oncological patients (35–100 years) and integrated these data with published single-cell brain RNA-seq atlases (700 patients) spanning the human lifespan.

Overall, transcriptomic differences between younger and older patients were modest compared with variation driven by spatial heterogeneity, epigenetic subtype (RTK1, RTK2, mesenchymal), driver mutations, and clinical factors such as preoperative dexamethasone treatment, underscoring the need for carefully controlled age-group comparisons.

Preliminary analyses indicate fewer OPC- and NPC-like malignant cells in older patients, accompanied by reduced non-malignant OPCs. In contrast, GPC-like tumor cells increased with age independently of EGFR amplification. Among non-malignant cells, astrocytes exhibited a more reactive A1-like phenotype, consistent with healthy aging, while microglia shifted from homeostatic to immunosuppressive transcriptional programs. Notably, age-related differences diminished toward the tumor core, where aging-associated signatures were most strongly expressed, suggesting convergence through tumor-driven activation of pathways normally upregulated during physiological aging.

To quantify aging-associated processes in GB relative to healthy controls, ongoing analyses integrate transcriptomic, epigenetic, and radiological aging clocks. Finally, we evaluate the clinical relevance of these processes by correlating them with survival and radiographic endpoints.

Together, this study provides a comprehensive view of how aging shapes the glioblastoma ecosystem and may guide age-adapted therapeutic strategies.



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